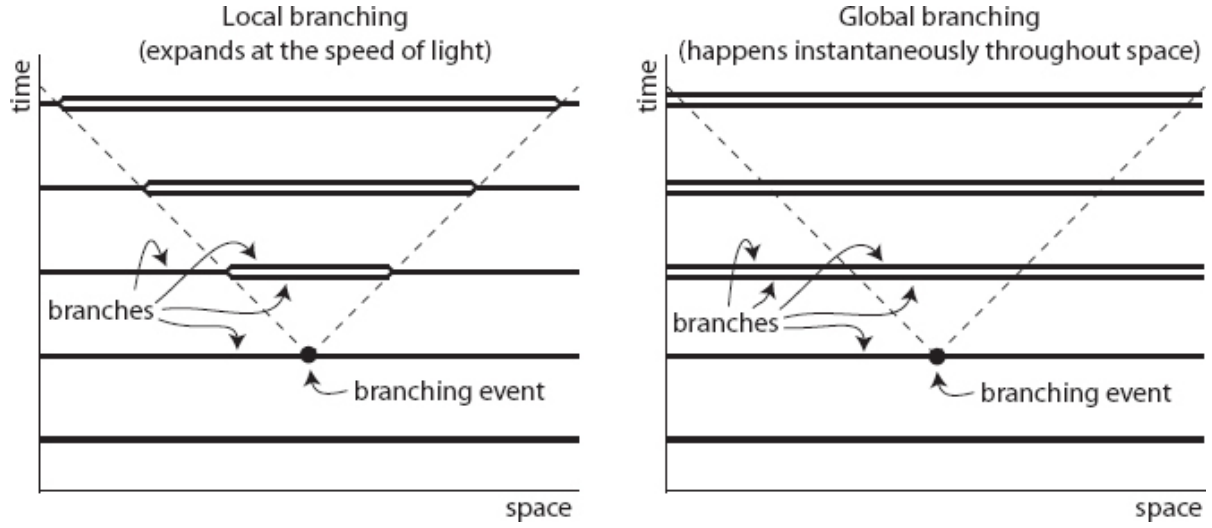
Her father studied the diagram. “I see. I guess I assumed that branching happened simultaneously throughout the universe, which bothered me as someone who is quite fond of special relativity. I’m sure you know as well as I do that different observers will define simultaneity differently. I kind of like this picture better, where branching propagates outward at the speed of light. All the effects look pretty local.”

Alice waved her hands before she resumed drawing. “But the other way works too. We are equally allowed to describe branching as happening all throughout the universe all at once. This view is helpful when we derive the Born rule using self-locating uncertainty, as we can sensibly talk about which branch you are on immediately after the branching occurs, no matter where it happened. Because of relativity, observers moving at different speeds will draw the branches differently, but there’s no observational difference caused by doing so.”



“Arrgh! You’ve just undone all of your good work. Now you’re telling me that branching can just as well be thought of as completely nonlocal.”

“Yeah, but what I’m actually saying is that the question ‘Is Many-Worlds a local theory?’ isn’t quite the right one to ask. It would be better to ask, ‘Can we describe branching as a local process, proceeding only inside the future light cone of an event?’ The answer is ‘Yes, but we can equally well describe it as a nonlocal process, occurring instantly throughout the universe.’”

Her father put his hands over his face, but he seemed to be trying to absorb this, not just giving up in frustration. Then he got up and mixed